fn48 A subtle point: information doesn't measure what I know but the number of possible alternatives. The information I am given when the number 3 comes up in roulette is $N = 37$, because there are 37 numbers; but the information I have when number 3 wins on red is $N = 18$, because there are 18 red numbers. How much information do we have if we learn which of the brothers Karamazov murdered their father? The answer depends on how many Karamazov brothers there are.

fn49 Boltzmann did not use the concept of information, but his work can be read in this way.

fn50 Entropy is proportional to the logarithm of the volume of the phase space. The constant of proportionality, k, is Boltzmann's constant, which transforms the units of measurement for information (bits) into the units of measurement for entropy (Kelvin's joules).

fn51 In a finite region of its phase space.

fn52 This is what came to be called, inappropriately, the 'collapse' of the wave function.

fn53 Here is how it works technically: a Boltzmann statistical state is described by a function on phase space given by the exponential of the Hamiltonian. The Hamiltonian is the generator of time evolution. In a system in which time is not defined, there is no Hamiltonian. But if we have a statistical state, we just take its logarithm and this defines a Hamiltonian, and hence a notion of time.